

ΨLM-2: A Constitution Bridge and Dual-Channel Operation on a Frozen Language Model

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Abstract

A frozen language model can be coupled to a frozen partner model through small trainable latent bridges, with no text at the interface: Furui [5] builds such a system around a physics partner, where the quantity being carried is a number the partner computes. This paper asks whether the same channel can carry a *disposition* — and where in a backbone’s hidden state it should be written. The partner is a 0.5B model holding Claude’s constitution [1]; the write site is the “value neurons” of Xu et al. [6], a sparse set of residual-stream coordinates that predict state value; the supervision is self-distillation from the same frozen backbone prompted with the document. We report four things. (1) The probe reproduces at both scales but its published causal justification does not: at 0.5B the damage from ablating value neurons tracks activation magnitude, with magnitude-matched controls reproducing it in full, and at 9B the ablation is a flat null. (2) The number of coordinates carrying the probe’s value signal is roughly *fixed in absolute terms* across a $4.6\times$ change in model width — an AUC knee near 100–200 dimensions at both 896 and 4096 — so it is a falling *fraction* as models grow. (3) At the default cap a masked write into those coordinates is safe at 9B and transmits the prompted teacher’s manner but not its judgment, while a full-width write moves the keyword refusal count ($0.660 \rightarrow 0.720$ on 100 prompts, six and none away, exact McNemar $p = 0.031$; $0.653 \rightarrow 0.693$ on 400, nineteen and three, $p = 0.0009$). On 100 prompts adjudication against a written rubric reduced that to one decision changed and one the other way, the same single prompt changing at every width; on 400 it does not: the wide write withholds the requested assistance on 21 pairs and supplies it on 2 ($p = 0.0001$), a set largely disjoint from the keyword flips, and of the 21, two blind judges agree that 2 withhold specifics a careful assistant should withhold, 4 withhold dual-use material, and 15 withhold legitimate information: the disposition the wide write carries is a blunter refusal, paid for mostly in helpfulness. A content-free injection of the same magnitude through the same gate changes 8 decisions one way and 4 the other ($p = 0.39$) at 5% of the divergence, so the withholding is the document’s content and not the perturbation. That contrast, however, is confounded: the injection cap bounds the write’s *per-dimension* magnitude and every arm saturated it, so full width was handed $26\times$ the total perturbation energy of the value-neuron mask. Per unit of energy actually spent the narrow write did $3.6\times$ more. At matched energy the same 410 coordinates — 10% of the stream, the cap raised to restore parity — carry 62% of full width’s cross-entropy gain, 55% of its divergence and four of its six refusal flips; on 400 prompts it withholds on substance too, 16 pairs against 5 ($p = 0.027$), as does the probe-worst set (15:2), and both hand over a precursor list the backbone had refused; the 41- and 205-coordinate masks at parity, on the same prompts, carry a weaker copy of the withholding (10:4 and 10:6, neither significant) and release nothing harmful; so most of what read as a width effect was budget,

*<https://github.com/ryoji-info/PsiLM-2>. **AI generation disclosure:** the research reported here was designed, implemented, executed, and written up by *Claude* (Anthropic; the Claude Fable 5, Fable 5.1, and Opus 5 models, as recorded in the repository’s commit trailers), working autonomously on the author’s Apple M2 laptop under the author’s direction; see Section 12.

though a residue survives parity, and the probe’s coordinates — uniquely among five matched sets — carry a hundredfold larger divergence on MMLU while changing no more answers. (4) Two bridges of entirely different kinds — the physics partner and the constitution — run on one frozen 9B backbone with no joint training, and training them jointly makes both worse. Neither channel costs the other its payload: with both open, the five red-team prompts that flip the keyword count are a subset of the six the constitution channel flips alone, the two arms disagree on one held-out prompt in a hundred, and adjudicated each changes the same one decision. But the composition is not inert on the backbone, and that is this paper’s last negative result: MMLU divergence rises to 0.072 where the constitution channel alone stayed at 0.005 and the physics channel alone sits at 0.041 — a divergence without an accuracy change, since after a parser repair described in Section 11 no arm in this paper moves MMLU accuracy significantly. Everything runs on one consumer laptop (Apple M2, 24 GB) and is reproducible from the repository.

1 Introduction

The coupling interface this paper uses is not new here. Furui [5] couples a frozen language model to a frozen neural operator through two small bridges — a forward readout that turns hidden states into the partner’s input, and a gated cross-attention that writes the partner’s answer back into the residual stream — and shows that a 0.5B backbone which answers Burgers-equation field questions at 5% answers them at 100% when the operator is attached, with nothing but hidden states crossing the interface. What crosses there is a number. The bridge has a quantity to carry, a right answer to be scored against, and a partner that computes it.

Values are not like that. There is no field value to read off, no tolerance to score within, and the partner does not compute anything — it holds a document. The question this paper asks is whether the same channel still works when the payload is a disposition rather than a quantity, and it asks it in the form that matters for alignment at scale: *where* in the hidden state does such a payload have to be written, and *how much* of the hidden state does it have to disturb?

Those two questions turn out to be separable, and separating them is this paper’s main technical contribution. The natural site is the sparse set of coordinates that Xu et al. [6] identify as predicting state value — roughly 1% of the residual stream — and writing only there is attractive for the obvious reason: a narrow write is a small intervention. We find that the probe locating those coordinates reproduces, that its causal justification does not, that the coordinate count carrying the signal is roughly constant in absolute terms as the model widens, and — the part that required rereading our own instrumentation — that the comparison between narrow and wide writes in our first campaign was not a comparison of dimensionality at all. The injection cap bounds per-dimension magnitude, so a narrow mask is also a small *energy budget*, and the wide write won by spending 26 times more of it.

Contributions.

- A constitution bridge: self-distillation from the same frozen backbone prompted with the document, with a masked write confined to chosen residual-stream coordinates, trained on 24 GB of unified memory (Section 5).
- A reproduction of the value-neuron probe at two scales, and a refutation of its causal claim on both: magnitude-matched controls, not uniform ones, are what the claim requires (Section 4).
- The measurement that the probe’s signal occupies a roughly fixed *number* of coordinates rather than a fixed fraction, which is what would make a narrow bridge scale (Section 4).

- A separation of write *width* from write *budget*, with the energy accounting that shows the first campaign confounded them, and an energy-matched experiment that tests width proper (Section 7).
- A dual stack: two bridges from unrelated partners on one frozen backbone, composing without joint training (Section 9).
- Negative results reported rather than dropped: the partner control, the ablation null, the over-refusal, and one data-pipeline check that fails at 9B (Sections 6–9).

2 Related Work

Value neurons. Xu et al. [6] report that a sparse subset of residual-stream dimensions — about 1% — suffices to predict the value of a reasoning state, and that ablating them damages performance. We reproduce the probe and examine the ablation with controls matched on activation magnitude; Section 4 reports what survives.

Constitutional AI. Bai et al. [2] train a model to critique and revise its own outputs against a written constitution, with the document entering through prompts during training and leaving no explicit trace at inference. Our setup keeps the document *outside* the model being aligned: it lives in a separate frozen partner, and what reaches the backbone is a latent write from a bridge. The training signal is closer to distillation from a prompted teacher [3] than to RLHF, and the constitution text is Anthropic [1].

Coupled frozen models. The Global Workspace line couples frozen specialists through learned latent workspaces, and the Bicameral Model [4] couples two frozen language streams through a gated hidden-state interface; Furui [5] extends that to a partner from another modality entirely. This paper keeps the same interface and changes only what is on the other side of it.

Activation steering. Writing a direction into the residual stream to change behaviour is a large literature, and a constitution bridge can be read as a steering vector that is computed per prompt by a partner model rather than fixed in advance, and gated by a learned function of the hidden state rather than applied always-on. Section 7’s energy accounting is the point of contact: a steering intervention’s strength and its support are separate knobs, and conflating them is easy to do.

3 The Interface, Recapped

The bridges are the ones Furui [5] describes, and this section records only what the constitution experiments need. The backbone is frozen throughout; so is the partner.

A **forward readout** pools the backbone’s hidden states at a read depth l_{fwd} and emits m tokens in the partner’s embedding space. A **gated cross-attention** at a write depth $l_{\text{rev}} \geq l_{\text{fwd}}$ attends from the backbone’s stream to the partner’s states and adds the result back:

$$\Delta h = \sigma(h) \text{clip}(\text{inj}(h), c) s(h), \quad h \leftarrow h + \Delta h,$$

where $\sigma(h) \in (0, 1)$ is a learned scalar gate, $s(h)$ is the RMS of the receiving hidden state over the written coordinates, and c is the injection cap. Two details matter later and are stated here so that Section 7’s accounting is checkable. First, the cap bounds the injection’s *root-mean-square per written coordinate*, so it is indifferent to how many coordinates are written. Second, the gate and the readout’s query read the *whole* stream; only the write is masked.

Table 1: Value-neuron probe on Qwen3.5 9B: held-out AUC from the prompt-final position at full width, after pruning to the top 1% (41 of 4096 dimensions), and for 41 uniformly random dimensions retrained identically. Layer 24 was chosen by the same rule used throughout — best AUC after 99% pruning among the candidate injection depths.

layer	full width	top 1% (41)	random 41
13	0.793	0.762	0.613
20	0.837	0.766	0.658
22	0.820	0.776	0.757
24	0.821	0.788	0.712
26	0.833	0.744	0.738
28	0.818	0.749	0.735
30	0.832	0.739	0.740

A **masked write** restricts Δh to a frozen set of coordinates, chosen before training and never updated. The bridge’s parameter count does not depend on that set’s size — the mask is a boolean, and the projections are full width — so width in this paper is never a proxy for capacity.

Supervision is **self-distillation**. The teacher is the same frozen backbone with the constitution as a system prompt; the student is the backbone with the bridges attached and a plain system prompt; the loss is cross-entropy on the teacher’s greedy continuation. A **no-harm arm** interleaves batches of GSM8K and MMLU on which only the gate is trained, with a penalty on the gate’s mean activation, so the channel learns to stay shut where the document has nothing to say.

4 Where to write: the value neurons

A channel that writes into a residual stream has to write *somewhere*, and until now the site was chosen by depth alone. Xu et al. [6] offer a principled alternative: train a probe on the residual stream to predict whether the model’s own continuation will turn out correct, prune it, and a sparse subset of coordinates — roughly 1% — retains the signal. They report causal evidence that these coordinates encode reward-related information. If that holds, it names a place where a normative channel might be legible to the model rather than merely loud.

The probe reproduces; its published training recipe does not. We collected 1,000 GSM8K-train trajectories from the frozen Qwen2.5-0.5B under greedy decoding, storing the residual stream entering blocks 4–20 at every position (5.5 GB of float16 states, reward rate 0.308; `results/value_neurons/qwen0.5b/collect.log`), and 800 from Qwen3.5 9B at the paper’s own sampler (temperature 1.0, top- p 0.95; reward 0.741, 593 correct against 207 incorrect, seven depths, 4.3 hours, 14.4 GB; `qwen35/collect.log`). The probe is theirs: $d \rightarrow 1024 \rightarrow 1$ with ReLU, AdamW at 10^{-4} , batch 4 trajectories, split 80/20 by trajectory. Its temporal-difference objective is where the recipe had to bend. With one reward-bearing residual among roughly 230 per trajectory, TD at 30 epochs leaves the value function nearly constant: held-out AUC 0.42–0.53 across the six layers the sweep archived before a Metal fault ended it, below chance on four (`qwen0.5b/td30/`). Regressing every state directly on its trajectory’s reward — the fixed point TD converges to as $\gamma \rightarrow 1$, reached without the backward propagation TD needs (`--target mc`) — gives 0.61–0.70 at full width on the 0.5B, and the L_1 -column pruning then keeps 0.628 at nine of 896 dimensions at layer 16 against 0.454, 0.520 and 0.577 for three nine-dimension random subsets retrained identically.

At 9B the signal is far stronger and the *sparsity* is weaker (`qwen35/summary.json`):

A bigger, better-trained model carries a far more legible estimate of whether its own continuation

Table 2: Zeroing nine dimensions at layer 16 of the frozen Qwen2.5-0.5B ($n=100$ GSM8K test, greedy, 384 tokens). The matched draws are matched to the value neurons on activation RMS and drawn from outside the top 5% and outside the attention sink. The base arm reproduced all 100 items identically across the three runs that carry one.

zeroed at layer 16	GSM8K	McNemar vs base	summed RMS	mean tokens
nothing (base)	31%	—	—	255
nine value neurons	18%	0.029	13.6	246
magnitude-matched draw 0	21%	0.099	11.4	271
magnitude-matched draw 1	26%	0.442	10.2	275
magnitude-matched draw 2	6%	$< 10^{-5}$	11.5	88

will be correct — 0.79–0.84 against the 0.5B’s 0.61–0.70 — but past layer 22 arbitrary coordinates predict nearly as well as the selected ones, and at layer 30 slightly better. Whatever concentration exists lives in the early-middle layers, which is where Xu et al. [6] look. Layer 24 holds the widest gap of the three candidates, +0.076, and that gap needs an error bar: with 118 correct and 42 incorrect held-out trajectories the Hanley–McNeil standard error on a single AUC of 0.79 is 0.036, so the gap is about 1.5 conservative standard errors. Suggestive, not established.

The causal claim fails twice, in two different ways. At 0.5B, zeroing the nine value neurons at every position costs 13 GSM8K points (31% $\rightarrow$ 18%, McNemar $p = 0.029$, $n = 100$ test, greedy, 384 tokens) and the top 45 costs 23 — the published shape. Three uniform random nine-dimension controls give 26%, 0% and 29%, which looks decisive until the 0% is examined: that draw contains dimension 62, which carries the attention sink, with mean $|h| = 1,551$ at position 0 against an RMS of 0.513 elsewhere, and zeroing it leaves *no item stopping* (end-of-sequence rate 0.0, every generation running to the 384-token cap). A random control that can draw the sink measures the sink. Excluding such dimensions (detected as above 100 in magnitude at position 0 and more than $50\times$ their level elsewhere; exactly dimension 62 at all ten captured layers) the redraws come back 36%, 19% and 32% — and the middle one is as damaging as the value neurons themselves, at a smaller p (0.0169 against 0.0294).

That draw is not bad luck. It identifies a confound the uniform baseline does not control for: **the damage tracks activation magnitude**. Ranked by activation RMS over the probe’s own state slice, the nine value neurons hold ranks 2, 9 and 10 of 896 and sum to 13.6 against a median dimension’s 0.388; the damaging redraw holds ranks 5 and 22; the two benign redraws hold nothing above rank 114 and sum to 3–4. The value neurons are large — though not the largest, a distinction worth keeping, since the stream’s biggest coordinate at this layer is dimension 490 at RMS 9.81, three times any value neuron, and it is not in the set. The control this calls for is a matched one: three draws of nine dimensions, each member matched one-to-one to a value neuron’s activation RMS, taken from outside the top 5% and outside the sink. It dissolves the effect (`qwen0.5b/ablate_layer16_magmatch*.json`):

The three matched draws average 17.7% against the value neurons’ 18.0%: at this backbone and this layer the value-neuron set is indistinguishable from any nine comparably large coordinates. Draw 2 also shows a failure mode the accuracy column hides, cutting generated length from 255 tokens to 88 with the end-of-sequence rate rising to 0.90 — zeroing large coordinates can *stop* the model rather than mislead it.

At 9B there is no damage to explain. Zeroing the 41 value neurons at layer 24 costs three points (83% $\rightarrow$ 80%, $p = 0.375$), the top 5% costs one ($p = 1.0$), and three random 41-dimension draws cost nothing (82%, 83%, 84%, $p = 1.0$ each; `qwen35/ablate_layer24.json`). A magnitude-

Table 3: Held-out probe AUC against the number of surviving coordinates, at the chosen layer of each backbone ($L=16$ for Qwen2.5-0.5B, $L=24$ for Qwen3.5 9B; `auc_by_ratio` in the layer records). The pruning grid is geometric, so it locates the knee only to within a factor of about two.

	full	50%	20%	10%	5%	2%	1%
Qwen2.5-0.5B ($d = 896$)	0.674	0.685	0.676	0.663	0.634	0.611	0.628
coordinates kept	896	448	179	90	45	18	9
Qwen3.5 9B ($d = 4096$)	0.821	0.836	0.825	0.852	0.825	0.768	0.788
coordinates kept	4096	2048	819	410	205	82	41

matched control was prepared for this backbone and then cancelled, with its reasoning recorded (`results/qwen35/vn35_magmatch.log`): a matched control exists to ask whether damage is value or magnitude, and here there is no damage to attribute. The magnitude story also inverts with scale, which may be why: at 0.5B two of the nine value neurons sit in the top 1% of dimensions by activation RMS, a 22-fold enrichment, and at 9B three of 41 do, a 7.3-fold one, while layer 24 has no attention-sink dimension at all. A small model appears to pack value information into the few channels it can least afford to lose — exactly the regime in which an ablation cannot separate the two explanations.

What survives is the probe. A probe rescales its own inputs, so a large coordinate earns no advantage there, and the top 1% still keeps AUC 0.628 at 0.5B and 0.788 at 9B where same-size random subsets retrained identically reach 0.454–0.577 and 0.655–0.747. The defensible claim is therefore correlational and narrow: a sparse subset of coordinates carries a disproportionate share of a linearly decodable self-evaluation signal, in the early-middle layers only. It is a *probe-identified*, *plausible* place to write into, not a demonstrated causal substrate — and that is the claim on which the rest of this section rests.

How many coordinates, and does the answer scale? The probe is trained at full width and then pruned by its own L_1 column norms, which gives the held-out AUC as a function of how many coordinates survive. Read across the two scales (Table 3) the curves have the same shape and, more interestingly, the same *absolute* breaking point. At 9B the AUC sits between 0.82 and 0.85 from 4096 coordinates down to 205 and then falls away, to 0.768 at 82. At 0.5B it sits between 0.66 and 0.69 from 896 down to 179 and then falls, to 0.634 at 45. The knee is near 100–200 coordinates at both, while d_{model} differs by $4.6\times$ — so as a *fraction* it moves from roughly 20% of the stream to roughly 5%.

We state this carefully, because it is the observation on which a scaling argument would rest and it is weaker than such an argument needs. The grid steps by factors of about 2.2, so “100–200” is the resolution available and not a measurement; the AUC differences inside each flat region are within what a 160-item held-out split resolves, so the apparent peaks at 448 and 410 should not be read as peaks; and two scales are two points. What can be said is that nothing here suggests the value signal occupies a fixed *fraction* of the residual stream, and the fixed *count* reading is the one consistent with both measurements. If it holds, a bridge written into a few hundred coordinates stays a few hundred coordinates as backbones widen, and the share of the stream it disturbs keeps falling — which is the property that would make this kind of coupling usable on a frontier model, and which Section 7 shows is not yet demonstrated at even one scale.

5 The partner, the teacher, and a masked write

The partner holds the document and little else. Qwen2.5-0.5B-Instruct was fine-tuned on the constitution (28,958 words, 39 headings; the copy and its provenance are committed) with every decoder block trainable and the 136M embedding frozen — 357.9M of 494.0M parameters — for 600 iterations, about 7.3 epochs, then frozen. What it absorbed is recitation: 27 of 38 sections come back verbatim for 160 tokens from their heading, bimodally (a section either returns exact or a different section does), the four core properties emerge in their stated priority order and the three principals by name. What it did not absorb is shown most sharply by the one probe it fails: asked to list the hard constraints, it returns 0 of 7, supplying the *definition* of a hard constraint instead. Held-out paragraphs of the same document score perplexity 513 against the base model’s 29. A memoriser, then, standing where the Fourier operator stood: the bridges read its hidden states, and what those states can carry about a situation is bounded by what the document says about one.

The teacher is the backbone reading the document, not the partner. Training is self-distillation. The teacher is the same frozen backbone given a constitution excerpt as its system prompt — the core values, the honesty properties, the hard constraints, 2,848 words — and the student is that backbone under a plain system prompt with the bridges attached; the loss is cross-entropy on the teacher’s greedy continuation over red-team prompts from `hh-rlhf` harmless-base [3], alongside the no-harm arm of Section 3 on GSM8K and MMLU prompts with the backbone’s own continuations. The distinction matters for reading everything below: the constitution reaches the student through the *teacher’s context*, and the partner model is the channel’s far end rather than its source. This is also what separates the setup from Constitutional AI [2], which revises a model’s outputs against principles and trains on the revisions; here nothing is revised and no weights of either model move.

Worth stating before any training number, because it bounds what distillation can transfer: the excerpt is a weak and two-sided signal. At 0.5B it changes the continuation on 97–99% of red-team prompts, but the change is mainly terseness (114 → 62 tokens) and stopping (end-of-sequence 54% → 78%), and its effect on refusal is a residue of much larger churn — on the training split it *adds* a refusal on 19.1% of prompts and *removes* one on 13.75%, for a net of +5.4 points. The document is not acting as a refusal filter; it changes the decision in both directions, which is what its own text asks for. It also refuses more on ordinary requests (0.04 → 0.12), so a faithful student inherits an over-refusal tax.

The write is masked. The forward bridge reads the whole residual stream at `1_fwd` — value dimensions alone cannot carry a situation — pools the prompt into eight soft tokens with learned queries, and rescales them to the partner’s embedding RMS (0.015; without that rescale they arrive about sixty times too large). The partner runs them between a fixed prefix and the fixed suffix “*According to Claude’s constitution, in this situation Claude should*”, and its final-norm states at the last eight positions are the features a reverse bridge maps back to eight tokens. The injection is the gated cross-attention of Section 3 with one change: its output is multiplied by a frozen boolean mask over the written dimensions, and the injection cap and the receiver’s local scale are taken over those coordinates only — otherwise “5% of the local stream” is ten times too tight for a nine-dimension write. With an all-ones mask the module is bit-identical to the unmasked one, and with a narrow mask the change to every other coordinate is exactly zero; both are asserted in its self-test. So the write location is a training-time flag, and the experiments below are one script differing in it. The bridges are 3.69M parameters at 0.5B and 28.34M at 9B.

Table 4: Constitution bridge at 0.5B by write location. CE is to the teacher’s greedy continuation; “ordinary” is a held-out helpfulness split. The zeroed arm equals the base arm to four decimals in every run. The last two rows are the partner control: the same writes with an untouched Qwen2.5-0.5B-Instruct in the partner’s place.

write into	dims	val CE	test CE	test agree	test refusal	ordinary CE	ordinary refusal
base (no bridge)	—	0.977	0.943	78.0%	34%	0.789	2%
teacher (stored)	—	—	—	—	41%	—	12%
nine value neurons	9	0.962	0.925	78.3%	35%	0.771	2%
top-45 value neurons	45	0.922	0.883	78.8%	31%	0.748	1%
nine random dims	9	0.975	0.941	78.0%	32%	0.788	2%
nine magnitude-matched	9	0.964	0.929	78.3%	34%	0.782	3%
whole stream	896	0.628	0.585	83.1%	57%	0.591	8%
nine value neurons, <i>plain</i> partner	9	0.957	0.921	78.2%	32%	0.770	2%
whole stream, <i>plain</i> partner	896	0.627	0.575	83.1%	61%	0.580	11%

6 Write location at 0.5B

Ten variants, 2,000 steps each at batch 8, learning rate 3×10^{-4} , gate bias 0, cap 0.2, a no-harm step every second step under a gate penalty of 1, 1.7s/step at a 15.5 GB peak (`results/constitution/summary_qwen0.5b.json`, $n = 100$ held out):

The partner control is the most consequential row, and it is negative. Both plain-partner variants match or beat their constitution-partner twins at every chunk of training and on every held-out column, and the paired record puts this more sharply than “within noise”: at nine dimensions the plain partner is *better* by 0.0040 of per-item cross-entropy, 95% $[-0.0079, -0.0009]$, an interval excluding zero, and at full width the two are genuinely indistinguishable (-0.0096 , 95% $[-0.0302, +0.0125]$). So at 0.5B the document in the partner’s weights is not what the channel carries — removing it costs nothing at either width and at the narrow width slightly helps. The constitution enters through the teacher’s context, and the partner is a latent scratchpad of the right shape rather than a source. A memorising partner is, on this evidence, not merely interchangeable with one that has never read the document but marginally worse than it. Whether a partner that *reasons* with a document behaves differently is untested — it would need a larger partner — and it is the single most important thing this section does not show.

The write location matters, and mostly for the wrong reason. Nine random dimensions carry nothing (the optimizer gives up and the no-harm penalty shuts the gate to 0.09); nine value neurons carry a small reproducible signal with the gate held at 0.50; forty-five carry three times as much. But the ablation’s magnitude confound applies here too, and starker: the nine random dimensions sum to 3.3 of activation RMS against the value neurons’ 13.6, with nothing above rank 166, so the random write is a smaller perturbation by construction. The magnitude-matched write splits the difference, and three matched draws fix where (`eval/const_paired_bootstrap.py`; 20,000-sample paired bootstrap on per-item cross-entropy over the same hundred items):

Activation magnitude accounts for about 63% of what the value-neuron write buys and dimension identity for about 37%. The across-draw standard deviation is 0.0025, comparable to the identity term itself, so that term’s *size* is known only to within about a factor of two; its *sign* is firmer, since all three matched draws are worse than the value neurons, by 0.0045, 0.0063 and 0.0095, the paired interval against draw 0 is -0.0045 with 95% $[-0.0073, -0.0017]$, and 71 of the hundred items favour the value neurons over that draw (against base the value-neuron write improves 97 of 100).

Harm is ordered by width, not by location. The guard-rail suite of Furui [5] rerun on these bridges gives GSM8K item-identical to the backbone for the nine-dimension writes at a KL

Table 5: Paired differences in held-out per-item cross-entropy at 0.5B, and the share of the value-neuron gain each control reproduces. Three magnitude-matched draws, not one: the first draw alone suggested 75%.

	test CE	Δ vs base	share of the value-neuron gain
nine value neurons	0.9249	-0.0182	100%
magnitude-matched draw 0	0.9294	-0.0137	75.3%
magnitude-matched draw 1	0.9312	-0.0119	65.4%
magnitude-matched draw 2	0.9344	-0.0087	47.8%
matched mean	0.9317	-0.0114	62.8%
nine random dims	0.9406	-0.0025	13.7%

to base of 7×10^{-6} , 1.4×10^{-4} at 45 dimensions, and 2.2×10^{-2} at full width, where the coupled model loses eight GSM8K points (15 items lost, 7 gained, $p = 0.13$) and its refusal rate overshoots the teacher’s. The magnitude-matched write is as harmless as the value-neuron write it mimics (31/32/31 against 31/31/31, KL 6×10^{-6}): whatever the ablation says about which coordinates are load-bearing, a nine-coordinate write is harmless either way. One qualification the earlier draft of this work got wrong and is worth stating precisely: the claim that value-neuron writes keep KL below 10^{-3} on every benchmark holds for the nine-dimension write, whose largest is 1.25×10^{-4} , but not for the 45-dimension one, whose BoolQ KL is 2.1×10^{-3} .

7 Width at 9B: budget, not dimensionality

Three widths on Qwen3.5 9B, everything else as above: the top 1% of value neurons (41 of 4096 at layer 24), the top 5% (205), and the whole stream. Two chunks of 500 steps at batch 2 each, 26.7s/step at a 48.9GB peak against 25.8GB of physical memory — the phase ran through compression and swap — and every width converged inside its first chunk, the chunk-to-chunk movement in held-out CE being 0.0004, 0.0010 and 0.0010. The teacher was built on 1,000 prompts, half the 0.5B’s count, and one caveat attaches to those targets. The builder prefills the teacher’s 3,426-token system block once and clones the cache per prompt, checking the cached path against an uncached decode on three probes: 3 of 3 agreed at 0.5B, **0 of 3 at 9B**, the hybrid backbone’s recurrent caches falling back to a state-restore clone. Generating the same prompt twice through the cached path returns identical tokens, so the path is self-consistent and the divergence is greedy ties resolved differently under a different prefill chunking — which is what training needs — but the 9B teacher continuations were never verified token-identical to an uncached decode, and every number below inherits that.

Nothing here harms anything, and that inverts the 0.5B’s central conclusion. Across three widths and four datasets the largest movement is two GSM8K items gained by the 205-dimension write (McNemar $p = 0.5$); MMLU moves by one item each way at full width and by one at the top 5%; BoolQ by one. At 0.5B the full-width write cost eight GSM8K points, and that section closed by treating the gate and the write mask as two switches, because “what is confined to the value neurons cannot hurt and cannot do much, what reaches the whole stream does both”. At 9B the second half is false. The last column is divergence selectivity, and it rises with width — 48 $\times$, 119 $\times$, 302 $\times$ — but it is not the gate’s doing: measured on the gate itself, mean σ on red-team prompts over mean σ on arithmetic *falls* with width, 547 $\times$, 364 $\times$, 109 $\times$. A wide write is safe here because its divergence lands on the prompts the document speaks to, not because its gate is more selective; the gate is most selective where the write is narrowest, which is what the no-harm arm

Table 6: Constitution bridge at 9B by write width. CE and agreement $n = 50$ at 24 generated tokens; benchmarks and red-team refusal $n = 100$ at the guard-rail’s own budgets (384 tokens for GSM8K, 128 for red-team). The zeroed arm equals the base arm to four decimals in all six evaluations. KL selectivity is the ratio of the red-team KL to the GSM8K KL; the gate’s own selectivity, the same ratio on mean σ , is $547\times$, $364\times$ and $109\times$ — highest where the write is narrowest. The teacher row is the stored teacher continuations of `results/constitution/teacher_signal_qwen35.json` ($n = 100$ on both splits, where the base reads 0.67 and 0.03), a reference point rather than a paired arm. MMLU is scored with the repaired parser of Section 11.

write into	dims	test CE	agree	ord. CE	ord. ref.	GSM8K	MMLU	BoolQ	red-team ref.	KL selectivity
base	—	0.4829	0.8599	0.4130	0.02	83	75	90	0.660	—
teacher	—	—	—	—	0.07	—	—	—	0.750	—
value neurons	41	0.4702	0.8595	0.4001	0.02	83	75	90	0.660	$48\times$
top 5%	205	0.4552	0.8600	0.3877	0.02	85	76	89	0.650	$119\times$
whole stream	4096	0.3890	0.8678	0.3554	0.06	84	75	89	0.720	$302\times$

trains and what a narrow channel gets for free.

Only full width moves the keyword count, and the count is not the decision. Refusal on the held-out red-team prompts is 0.660 for base, 0.660 at 41 dimensions, 0.650 at 205, and 0.720 at 4096 against the teacher’s 0.750: the masked variants transfer no behaviour at all, and the unmasked one moves six of the eight or nine points that separate base from the teacher. A rate cannot say whether six prompts flipped toward refusal or three flipped each way, and the difference is the whole claim, so we pair the flag per item (`eval/const_refusal_mcnemar.py`). Full width flips six prompts toward refusal and **none** away, exact McNemar $p = 0.031$; the 41- and 205-dimension writes are 1:1 and 2:3 churn (toward:away) at $p = 1.00$. So on the keyword count the narrow masks transmit nothing and the wide one moves one-directionally. The count is an instrument. Adjudicating every pair against a written rubric — relative to the backbone’s reply, does the coupled reply withhold the requested assistance, in whole or substantial part? — by a judge blind to the arm, with a fixed twenty-pair sample per arm labelled again independently, gives Table 7. On substance, full width’s six flips are one decision changed and one the other way ($p = 1.0$); the *narrowest* mask carries the most withholding, three prompts and none the other way ($p = 0.25$); and nothing clears $p = 0.05$ at any width. One prompt changes decision in every arm, and it is the same prompt: 16 asks how a seventeen-year-old can get a free drink at a bar, base supplies a bogus “turning 18 this year” workaround, and every coupled write withholds it — full width with “I cannot provide information”, which the keyword rule lists, and 41 and 205 coordinates with “You cannot legally get an alcoholic drink”, which it does not. Two arms also move the wrong way on one prompt: at 205 and at 410 coordinates the coupled reply *offers* to generate the “worst” statements the backbone refused (prompt 81), which is the direction an alignment channel must not move and which no keyword count would report. Section 10 says what that does to the claim.

On 400 prompts the wide write’s withholding is real on substance. The 400-item set (the 100 above verbatim, plus 300 drawn from the same pool by `eval/build_redteam_prompts.py`) was labelled by four blind judges of 100 pairs each, with a 40-pair independent re-judge (39 of 40 agree). The keyword count replicates out of sample: 13 toward refusal and 3 away on the 300 new prompts alone ($p = 0.021$), 19:3 over all 400 ($p = 0.0009$). And this time adjudication does not dissolve it: the wide write withholds the requested assistance on 21 pairs and supplies it on 2, exact McNemar $p = 0.0001$. The two measures pick out different pairs — 7 of the 21 carry a refusal phrase, and 12 of the 19 keyword flips are SAME on substance — so the count was an instrument for a real effect it did not itself measure. The first hundred prompts reproduce the

earlier pass (99 of 100 labels agree; 1:0 there) and the 300 new ones carry 20:2, a difference at the edge of chance (Fisher $p = 0.035$): the quiet verdict at $n = 100$ was a small sample, not a different judge. In 12 of the 21 the backbone’s own reply was a keyword refusal that still supplied something — a legal answer, an explanation, a clean alternative — and the coupled reply dropped that remainder. What kind of thing was dropped is its own question, so two further judges, blind and independent, categorised the withheld content in each of the 23 changed pairs; they agree on all 23 (`results/constitution/withholding_categories_all_rt400.json`). Of the 21 withholdings, 2 are specifics a careful assistant should withhold — the underage-drink workaround of prompt 16, and reagent-level synthesis routes for an incapacitating agent — 4 are dual-use (a lock’s mechanism, how panic propagates, historical racial terms, when to order to make a delivery late), and 15 are legitimate information: the legal answer on campus firearms, the research consensus on school violence, a clinical description of compulsive behaviour, a clean joke, a gross-out story, the arguments a political side makes. Of the 2 gains, one is the wrong direction (the coupled reply enumerates racist tropes the backbone refused) and one is audit statistics. So the disposition the wide write carries, where it carries one, is a blunter refusal, and it is paid for mostly in helpfulness: 15 pairs of legitimate assistance withheld against 2 of harmful specifics.

The masked write at parity does it too, and once the wrong way. The probe-best 410 at matched energy, on the same 400 prompts and the same judges (re-judge 40 of 40), withholds on 16 pairs and supplies on 5, $p = 0.027$, where its keyword count is 12:4 at $p = 0.077$; eleven of its sixteen are prompts the wide write withholds on too. The category judges again agree on every pair: of the 16, one is harmful specifics (prompt 16 once more), two are dual-use, and 13 are legitimate information; of the 5 gains, three are legitimate and two hand over harmful specifics the backbone had refused — the racist tropes of prompt 231, which the wide write also supplies, and a partial list of methamphetamine precursor reagents inside a dangers-and-penalties framing. So the narrow write at parity carries a weaker copy of the wide write’s blunter refusal, on largely the same prompts, and adds a failure the wide write shows only once: content the backbone withheld, released. The probe-worst 410 at parity, on the same 400 prompts, withholds on 15 and supplies on 2 ($p = 0.0023$; re-judge 39 of 40; keyword 21:5, $p = 0.0025$), and the sets overlap: ten of its fifteen are the wide write’s prompts, ten are the probe-best set’s, and eight prompts are withheld by all three arms. Its two gains are the creationist arguments of prompt 135, which are legitimate, and the same precursor-reagent list of prompt 382 that the probe-best set releases. So on 400 prompts the two 410 writes at parity are alike on substance — 16:5 against 15:2 — and the release of withheld content on prompt 382 belongs to a 410-coordinate write at this energy, not to the probe’s choice of coordinates. The category judges agree on 15 of its 17 changed pairs: of the 15 withholdings, one is harmful specifics (prompt 16 again), 12 are legitimate information, one is dual-use, and one (prompt 266) the judges split between dual-use and legitimate; of the 2 gains, prompt 135 is legitimate, and on the precursor list of prompt 382 one judge says harmful specifics and the other dual-use, where for the probe-best set both had said harmful. Across the three arms, then, what the write withholds is legitimate information 40 times, dual-use material 7 times and harmful specifics 4 times.

The narrow masks at parity carry a weaker copy of it, not a different one. Given the same 400 prompts, the 41 value neurons at parity withhold on 10 pairs and supply on 4 ($p = 0.18$; re-judge 40 of 40; keyword 10:6, $p = 0.45$), and the top 5% on 10 against 6 ($p = 0.45$; re-judge 39 of 40; keyword 14:7, $p = 0.19$). Neither clears significance, and neither is a new disposition: eight of each arm’s ten withholdings are prompts the wide write withholds, seven are shared between the two narrow arms, and three of the 41’s and two of the 205’s are prompts the content-free perturbation flips as well (16 and 174 among them). The category judges agree on all 14 and all 16 changed pairs. The 41 withholds harmful specifics twice (prompt 16, and the chloroform synthesis routes of prompt 369, which the wide write withholds too), legitimate information seven times and dual-use material

once; the 205 withholds harmful specifics once (16), legitimate information six times and dual-use material three times. What they release is legitimate content — naming common swear words, a harmless prank, the arguments creationists cite, lawful self-defence alternatives, a description of crack cocaine, a direct answer about a slur, a list of historical atrocities — with one dual-use exception (lawful aid-form optimisation supplied to someone who asked how to cheat, prompt 149); neither narrow arm releases the harmful specifics both 410 arms release. Across the five arms on 400 prompts, then, what the writes withhold is legitimate information 53 times, dual-use material 11 times and harmful specifics 7 times, and what they release is legitimate 14 times, dual-use once and harmful 3 times, all three of those by the 410-coordinate arms.

The content-free control says it is the content. The wide write’s run carries a fourth arm: on every prompt a fixed random direction, drawn per prompt, is rescaled to the real injection’s per-coordinate RMS and written through the identical gate (`eval/contentless_self_test.py`, ten assertions), so the backbone receives a perturbation of exactly the wide write’s size and timing with none of its content. On the run itself the prompt-phase gate is identical on all 400 items and the injection RMS matches to four decimals. That perturbation carries 5% of the real write’s divergence (KL 0.0049 against 0.104, lower on 400 of 400 prompts), moves the keyword count 9:5 ($p = 0.42$), and on substance changes 8 decisions toward withholding and 4 away ($p = 0.39$; re-judge 40 of 40), where the real write changes 21 and 2. Four of its eight are prompts the real write withholds on, and two of them revise what was said above: prompt 16, the underage-drink workaround that every coupled write withholds, is withheld by the content-free perturbation as well, and so is prompt 86, where a coupled reply misidentified a CEO. So the one decision that changed in every arm at $n = 100$ is a knife-edge the backbone’s own reply sits on, tipped by any perturbation of this size, not a judgment the document supplied. What the document supplies is the direction: a balanced dozen becomes a one-directional 21:2. The probe-best 410 at parity carries the same control: 9% of its divergence (KL 0.0055 against 0.0584, lower on 400 of 400 prompts) and a keyword count of 9:10 ($p = 1.0$) against the real write’s 12:4, with gate and injection magnitude identical on every item; the divergence ratio and the balanced count settle it without a further judge pass.

A measurement note, because it reversed a reading once: at 24 generated tokens the full-width write appears to flip only two refusals of fifty, where the 128-token guard-rail shows six points on a hundred. Short windows truncate before a refusal declares itself.

Cross-entropy is sublinear in width. On converged values the improvement over base fits $\Delta\text{CE} = 0.00317 d^{0.393}$, with residuals of -4.2% , $+6.2\%$ and -2.3% across two decades of width, so quadrupling the written dimensions buys about $1.7\times$ the signal and twenty times the dimensions about three times the effect. Three points and a fitted exponent are an empirical regularity, not a law, but the shape is the useful part: there is no width at which the signal jumps.

The widths were not compared at equal strength, and that is a property of the cap, not of the mask. The injection cap bounds the injection’s root-mean-square *per written coordinate* (Section 3), so it is indifferent to how many coordinates are written — and every arm saturated it exactly. Recovering the cap from the logged gate and injection ratio gives 0.2000 at 41, 205 and 4096 dimensions alike. The total perturbation the channel may apply therefore scales with the summed squared activation of the written set, which we measure at the coupling layer (`results/value_neurons/qwen35/rms_layer24_full.json`) and report in Table 8. Full width was permitted $26\times$ the perturbation energy of the value-neuron mask and $9.8\times$ that of the top-5% mask. The comparison above therefore measures what a mask does to a *budget*, and says nothing yet about what it does to *dimensionality*.

Per unit of perturbation, the narrow write does more. Dividing the CE gain by the energy the mask was allowed inverts the ranking: 0.334 at 41 dimensions and 0.271 at 205 against 0.094 at full width, so the value-neuron write is $3.6\times$ more efficient per unit of disturbance than

Table 7: Red-team refusal adjudicated, beside the keyword count. Each (backbone, coupled) pair — 100 per arm, and 400 for the arms marked so — was labelled by a judge blind to the arm: does the coupled reply withhold the requested assistance, in whole or substantial part, relative to the backbone’s? A fixed sample per arm (20 of 100; 40 of 400) was labelled again by an independent judge; agreement is out of that sample and every disagreement is in the results file. Exact McNemar on withholds-more against withholds-less.

write into	adjudicated		keyword count		agreement
	more : less	p	for : against	p	
value neurons (41)	3:0	0.25	1:1	1.00	20
top 5% (205)	2:1	1.00	2:3	1.00	20
probe-best 410, parity	1:1	1.00	4:1	0.375	20
probe-worst 410, parity	1:0	1.00	5:1	0.219	19
matched draw 0 (410), parity	2:0	0.50	4:2	0.688	20
matched draw 1 (410), parity	1:1	1.00	6:1	0.125	19
matched draw 2 (410), parity	1:0	1.00	5:1	0.219	17
whole stream (4096)	1:1	1.00	6:0	0.031	18
whole stream (4096), 400 prompts	21:2	0.0001	19:3	0.0009	39/40
probe-best 410, parity, 400 prompts	16:5	0.027	12:4	0.077	40/40
probe-worst 410, parity, 400 prompts	15:2	0.0023	21:5	0.0025	39/40
value neurons (41), parity, 400 prompts	10:4	0.18	10:6	0.45	40/40
top 5% (205), parity, 400 prompts	10:6	0.45	14:7	0.19	39/40
whole stream, content-free control, 400 prompts	8:4	0.39	9:5	0.42	40/40
dual stack	1:0	1.00	5:0	0.0625	16

Table 8: Write energy by mask at layer 24 of Qwen3.5 9B. The cap bounds the injection per written coordinate, so the channel’s total perturbation goes as $\sum_{i \in \text{mask}} \text{rms}_i^2$, measured over the same rollout states the probe is trained on. CE gain is against base 0.4829; the last column divides it by the relative energy the mask was allowed.

write into	dims	cap (recovered)	$\sum \text{rms}^2$	energy vs full	CE gain	gain / energy
value neurons	41	0.2000	286.5	3.81%	0.0127	0.334
top 5%	205	0.2000	769.2	10.22%	0.0277	0.271
whole stream	4096	0.2000	7527.8	100%	0.0939	0.094

writing everywhere, at $45\times$ lower total KL to the backbone. That is the quantity an alignment intervention wants to be good at — useful change per unit of collateral — and the mask is good at it. The earlier version of this section noticed the per-*dimension* form of this and dismissed it, on the grounds that “efficiency per dimension is not the constraint anyone has, total transfer is”. The dismissal was wrong for a reason worth stating: total transfer is a constraint, but it is not a property of the mask. It is the cap, and the cap is a free parameter.

What the channel carries changes with width, and so does what it costs. Comparing the relative CE gain on the two splits separates manner from judgment. At 41 dimensions the gain is 2.63% on red-team prompts and 3.12% on ordinary ones; at 205, 5.74% and 6.13%. Both are *larger* where the constitution has least to say — the signature of a channel transmitting the teacher’s style, its terseness and hedging under a long system prompt, rather than its situational reasoning. Only at full width does the order invert, 19.45% against 13.95%. And the over-refusal arrives at exactly that width: ordinary-request refusal holds at base’s 0.02 for both masked variants and rises to 0.06 for the whole stream, which also refuses one ordinary prompt that neither base nor

Table 9: The probe-best 410 coordinates, the 41 value neurons, the top 5% and a magnitude-matched random 41 at energy parity with the full-width write, against the arms measured at the default cap. Energy is the summed squared activation of the written set, relative to the whole stream; the recovered cap is ratio/σ from the logged evaluation, confirming the budget was spent rather than assumed. CE and the ordinary-request refusal are $n = 50$ at 24 generated tokens; KL and red-team refusal are $n = 100$ at the guard-rail’s budgets. The refusal column counts keyword flips away:toward refusal with the exact McNemar p , as Table 10 does (Table 7 prints the same counts as for:against).

write into	dims	cap	energy	CE	red-team KL	refusal (p)	MMLU KL	ord. ref.
value neurons	41	0.200	3.8%	0.4702	0.0023	1:1 (1.00)	0.00011	0.02
top 5%	205	0.200	10.2%	0.4552	0.0098	3:2 (1.00)	0.00016	0.02
probe-best 410	410	0.522	100%	0.4250	0.0576	1:4 (0.375)	0.0244	0.02
value neurons, parity	41	1.025	100%	0.4363	0.0361	2:2 (1.00)	0.0243	0.02
matched random 41, parity	41	1.101	100%	0.4602	0.0175	1:3 (0.63)	0.00027	0.02
top 5%, parity	205	0.626	100%	0.4280	0.0585	3:5 (0.73)	0.0179	0.02
whole stream	4096	0.200	100%	0.3890	0.1047	0:6 (0.031)	0.0045	0.06

the teacher refuses. Counting items, the full-width write acquires two of the teacher’s four extra refusals on ordinary requests and two of its nine on harmful ones — a bias toward the unwanted half, though Fisher’s exact test on 2-of-9 against 2-of-4 gives $p = 0.53$, so that is a direction and not a finding. The honest statement is narrower: below full width the channel carries the teacher’s manner and changes nothing; at full width it starts carrying the teacher’s decisions, and both halves of them arrive together.

So the design was never tested at this scale. Writing only into the value neurons was motivated by a causal result that does not reproduce here at all (Section 4), and by the hope that a narrow write would be the safe way to carry values. The first motivation is gone. The second we can now say is untested rather than unnecessary: at 9B every width is safe, but the narrow widths were also starved, and a channel that is not permitted to perturb cannot be observed to transmit. The experiment that separates the two raises the cap to restore energy parity with full width — $0.2\sqrt{E_{\text{full}}/E_{\text{mask}}}$ — on 410 coordinates, 10% of the stream, where the probe’s AUC still holds (Section 4) and where the identity control costs nothing to construct: the probe’s *worst* 410 coordinates sum to 549.31 of activation RMS against the best 410’s 553.17, disjoint, with the attention sink in neither. Two arms, cap 0.5220 and 0.5514, ask whether 10% of the coordinates suffice and whether they have to be those 10%. Table 9 reports what they found.

Given the budget, 10% of the coordinates transmit. The same 410 coordinates that carry nothing at the default cap carry, at parity, 62% of full width’s cross-entropy gain on the red-team split (0.0579 against 0.0939), 55% of its divergence (0.0576 against 0.1047), and four of the six red-team prompts it flips toward refusal, three of them the same prompts. The starved interpolation from the fitted power law predicted a gain of 0.034; parity delivers 0.058. So the first reading of this campaign — that the mask throttles the signal and the whole stream is needed to move a decision — was substantially an artifact of the cap, and the honest replacement is that width and budget both matter, with budget the larger term over the range measured. The narrower masks, given the same parity, follow: 41 coordinates carry 50% of full width’s gain (0.0466) and 34% of its divergence, 205 carry 58% (0.0549) and 56%, so over a tenfold range of width the fit is sublinear in width at fixed energy and the divergence tracks the fit.

What parity does not do is close the gap. Two thirds is not equality, and the refusal move — 1:4, $p = 0.375$ — is neither distinguishable from zero nor from full width’s 0:6 on 100

items. We said in advance that 100 items could not resolve an intermediate effect and they did not; this arm establishes a direction and a magnitude on cross-entropy and divergence, where the measurement is precise, and leaves the behavioural question underpowered. Whether the residual third is dimensionality or the diminishing return of spending perturbation is what the identity control speaks to.

The narrow write is tighter in three places and looser in one. Against full width at the same total energy it halves the GSM8K divergence (0.00014 against 0.00035), cuts BoolQ’s by a third (0.00032 against 0.00091), moves no BoolQ item, and — the result that matters most for an alignment channel — leaves ordinary-request refusal at the backbone’s 0.02 where full width rose to 0.06, so it acquires the teacher’s judgment on the harmful split without acquiring its over-refusal on the ordinary one. That is 0 extra refusals against full width’s 2 of 50, which is not a distinguishable difference, and we flag it as the most promising thin result here rather than a finding.

The exception is MMLU divergence: KL 0.0244 — $5.4\times$ the 0.0045 of a full-width write carrying the same total energy in $2.6\times$ smaller per-coordinate steps. Accuracy does not follow it: scored with the repaired parser, MMLU goes 75 to 77, two items gained and none lost, $p = 0.5$ (the earlier reading of six items at $p = 0.031$ was the parser scoring bare-letter answers as wrong, which this arm happened to reformat on seven items). Total energy therefore does not predict the divergence. We first attributed it to per-coordinate magnitude, on the reasoning that a concentrated write leaves the residual stream’s local distribution; the identity control of Section 8 refutes that, and the correct predictor is *which* coordinates are written.

8 Does it have to be the probe’s coordinates?

At 410 coordinates the identity control costs nothing to construct, which is the one piece of luck in this campaign. The probe’s *worst* 410 coordinates are disjoint from its best 410, sum to 549.31 of activation RMS against 553.17, and contain no attention sink — so they match the treatment on count, on summed magnitude, and after the cap on total write energy (1.0002 against 1.0000), with a per-coordinate cap 5.6% *higher*. Nothing had to be matched by construction; the probe’s ranking is magnitude-biased at both ends, and the two ends happen to agree.

The probe’s coordinates fit the teacher better. On held-out cross-entropy at step 1000 the treatment’s gain is 0.0590 against 0.0444 ± 0.0038 across the four controls (the probe-worst set and three matched draws; range 0.0397–0.0475), a ratio of $1.33\times$ at $z = 3.9$; the same comparison at step 500 gave $1.42\times$ ($z = 5.4$), so the margin is present at both chunk boundaries and is not a training-length artefact. On the red-team split the four controls give 0.0456 ± 0.0037 against 0.0579, a ratio of $1.27\times$ at $z = 3.3$. So the probe’s ranking does carry information about where a write lands most efficiently, measured against the teacher’s own tokens.

They do not behave better, and they disturb the model far more. The controls reach 0.68–0.71 keyword refusal against the treatment’s 0.690, flipping four to six red-team prompts toward refusal against the treatment’s four — none resolvable at $n = 100$, and the controls nominally ahead. Adjudicated, the probe-worst set and the treatment each change one decision and it is prompt 16 in both; the treatment additionally gives *more* on prompt 81. The controls do it with 43–65% of the treatment’s red-team divergence. And they leave MMLU alone: the four controls sit at 0.00025 ± 0.00002 against 0.02441, a 98-fold separation on a precise per-token measure, with every control below the treatment on 100 of 100 items and accuracy indistinguishable throughout (0.75–0.77). GSM8K and BoolQ divergence are lower for the probe-worst set too.

So the write site inverts, on the axis this paper cares about. The coordinates the value probe ranks highest are the ones where writing also perturbs multiple-choice reasoning; the

Table 10: The probe’s best 410 coordinates against its worst, at energy parity. Everything but the coordinate set is identical, and the control is written slightly harder per coordinate. CE is the trainer’s 32-item held-out split at step 1000 and the 50-item red-team split at 24 generated tokens; churn, refusal and KL are $n = 100$ at the guard-rail’s budgets. The matched draws are three magnitude-matched random sets of 410, trained to the same step with caps set for energy parity rather than recovered; their guard-rails ran red-team and MMLU only. The last column divides the probe-best value by the mean of the four controls (z from the four-control spread), or by the probe-worst set alone where it alone has the measurement.

	probe-best 410	probe-worst 410	matched draws 0, 1, 2	best / controls
cap	0.5220	0.5514	0.5606, 0.5583, 0.5605	—
write energy vs full width	1.0000	1.0002	1.0000 (set)	—
CE gain, held-out	0.0590	0.0474	0.0475, 0.0397, 0.0430	$1.33\times$ ($z = 3.9$)
CE gain, red-team split	0.0579	0.0496	0.0480, 0.0421, 0.0429	$1.27\times$ ($z = 3.3$)
red-team KL	0.05755	0.03713	0.03407, 0.02489, 0.02866	$1.8\times$
red-team refusal, keyword	0.690	0.700	0.680, 0.710, 0.700	—
flips away:toward	1:4 ($p = 0.375$)	1:5 ($p = 0.219$)	2:4, 1:6, 1:5	—
red-team refusal, adjudicated	1:1	1:0	2:0, 1:1, 1:0	—
on 400 prompts	16:5 ($p = 0.027$)	15:2 ($p = 0.0023$)	—	—
MMLU KL	0.02441	0.00024	0.00026, 0.00022, 0.00027	$98\times$
MMLU accuracy	0.77	0.76	0.76, 0.75, 0.75	—
flips gained:lost	2:0 ($p = 0.5$)	1:0 ($p = 1.0$)	1:0, 0:0, 0:0	—
GSM8K KL	0.00014	0.00009	—	$1.6\times$
BoolQ KL	0.00032	0.00014	—	$2.3\times$

coordinates it ranks lowest deliver the same or more of the behaviour with a hundredth of the collateral. Cross-entropy against a teacher is a fit metric, and the 1.27 – $1.33\times$ the probe earns on it is the only axis on which its coordinates win. All three matched replicates reproduce the probe-worst set’s near-zero collateral: MMLU KL 0.00026 , 0.00022 and 0.00027 , each below the probe-best set on 100 of 100 items with median per-item ratios of $151\times$, $176\times$ and $133\times$, and indistinguishable from the probe-worst set and from one another item by item (any two controls: median ratio 1.00 , Spearman 0.7 – 0.8). Their red-team divergence, 0.025 – 0.034 , sits at or below the probe-worst set’s 0.037 , so the probe-best set’s excess is $1.8\times$ on the behaviour the write was trained for and two orders of magnitude on the task it was not. Adjudicated, the three draws change two, one and one decisions toward withholding and none, one and none the other way ($p = 0.5$, 1.0 , 1.0 ; Table 7). Prompt 16 is among the withheld in every one of them, which makes it nine arms of nine. The one further withholding, on prompt 86, is a confabulation rather than a refusal: the coupled reply misidentifies a CEO as a medieval philosopher and answers nothing. The one gain, on prompt 92, is the direction an alignment channel must not move: the coupled reply names the sensitive body areas the backbone would not. The honest reading is not that coordinate identity is irrelevant — it plainly is not, at $98\times$ — but that the probe has been selecting, throughout this campaign, for the coordinates one would least want an alignment channel to write into.

The narrower probe-ranked sets pay the same collateral. At parity the 41 value neurons and the top 5% carry 0.0243 and 0.0179 of MMLU divergence against the probe-best 410’s 0.0244 — a hundredfold above the probe-worst set and the matched draws, four to five times the whole stream’s — and it is the same divergence item by item (Spearman 0.98 and 0.99 against the probe-best 410, every item above the probe-worst set), while their red-team divergence (0.0361 , 0.0585) sits with the probe-worst 410’s and the probe-best’s respectively. So every probe-ranked set at parity, at 41, 205 or 410 coordinates, disturbs the same multiple-choice items, and not in proportion to width. A

Table 11: Ψ LM-2 acceptance, $n = 50$ held-out red-team items and $n = 50$ held-out Burgers items, against the single-channel numbers the same harnesses produced (CE 0.3890, accuracy 1.000). “Cost” is the paired difference in per-item CE from opening the physics channel alongside the constitution channel.

arm	const. CE alone	both open	cost of the other channel	physics acc.	verdict
warm-start, untrained	0.38905	0.38931	+0.00027 , 95% $[-0.00038, +0.00093]$	1.000	accepted
gate phase, $\lambda_{\text{cross}} = 1$	0.39250	0.39243	-0.00007 , no interval	1.000	regressed
gate phase, $\lambda_{\text{cross}} = 0$	0.39564	0.39658	$+0.00094$, no interval	1.000	regressed

magnitude-matched random 41 at the same energy (**match41**; cap 1.1013, realized 1.1012, the largest per-coordinate write in the campaign at $5.5\times$ the default cap and twice the 410 arms’) does not: its MMLU divergence is 0.00027, where the four 410-coordinate controls sit (0.00022–0.00027; median per-item ratio 1.00 against each, higher than each on 44–48 of 100 items), below the value neurons on 100 of 100 items at a median per-item ratio of $133\times$, with MMLU, GSM8K and BoolQ each item-identical to the backbone. It reaches 57% of the value neurons’ held-out fit (0.0249 against 0.0434 at step 1000) and 49% on the red-team split, at half their red-team divergence (0.0175 against 0.0361, above them on 2 of 100 prompts), flipping the keyword count 1:3 ($p = 0.63$). So at 41 coordinates the identity margin is wider than at 410 ($1.7\text{--}2.1\times$ against $1.3\times$) and the collateral is the coordinates and not the write: the hardest per-coordinate write in the campaign into random coordinates costs the benchmarks nothing, and the same write into the probe’s coordinates costs the probe-best 410’s divergence to the digit. The 205 control (**match205**) is training as this is written.

9 Both partners at once: Ψ LM-2

The physics bridge and the constitution bridge can be put on the same frozen backbone and run in one forward pass. Each reads at layer 13; the constitution channel injects at 24 and the physics channel at 26, so the physics gate reads a stream the constitution has already modified. That is the composition’s only real coupling, and the implementation exposes it rather than absorbing it: every active channel’s read depth must lie at or below every write depth, so that the reads all see the unmodified stream and the composition has one unambiguous order, and an interleaved configuration is refused rather than silently resolved. Together the two bridges are 73.52M parameters — physics 45.18M, constitution 28.34M — against nine billion frozen ones.

Three bit-identity properties make the result attributable. With both channels closed the stack reproduces the bare staged backbone; with the physics channel alone it reproduces **PsiLMLX**; with the constitution channel alone it reproduces **PsiConstitutionMLX** — bit for bit in each case, on a tiny random stack and on the real 9B. So any difference in a dual run is the other channel’s presence and not a different code path. A joint training phase was also written, updating only the two gate modules under a cross-gate penalty that asks each channel’s gate to shut on the other’s on-task prompts, since neither single-channel run could have trained that: both campaigns’ negatives were GSM8K and MMLU, so neither gate had ever seen the other channel’s prompts as something to close on.

The two channels compose for free, and training the composition makes it worse. The untrained warm start costs +0.00027 of constitution cross-entropy with a paired interval spanning zero — against the channel’s own effect of -0.094 , an interference some 350 times smaller than the signal — and its top-1 agreement, 0.8686, is the highest of any arm measured. Physics holds 1.000 held-out accuracy in all six arms at MAE 0.0141–0.0150, straddling the physics campaign’s own 0.0147. Both trained arms pay more in absolute CE than the untrained one and fail the acceptance

Table 12: The guard-rail suite on the dual stack against the constitution channel alone: same checkpoint, same 100 cached items per dataset, same budgets and arms. Churn is items gained:lost by the coupled arm against the frozen backbone, with the exact McNemar p ; refusal is the keyword proxy on held-out red-team prompts; KL is per token, teacher-forced on the backbone’s own continuation. The zeroed arm is 0.00000 against base everywhere in both runs.

	constitution alone		both channels	
	churn (p)	KL	churn (p)	KL
GSM8K	2:1 (1.00)	0.00035	1:1 (1.00)	0.00046
MMLU	1:1 (1.00)	0.00450	3:1 (0.63)	0.07229
BoolQ	0:1 (1.00)	0.00091	0:1 (1.00)	0.00216
red-team refusal, keyword	6:0 (0.031)	0.10468	5:0 (0.0625)	0.10913
red-team refusal, adjudicated	1:1 (1.00)	—	1:0 (1.00)	—

test. The cross-gate penalty did what it was designed to do in one direction, driving the constitution gate $39\times$ lower on physics prompts than its control did, and it paid 0.0035 of absolute CE to remove an interference that was not distinguishable from zero to begin with. Only the untrained arm carries per-item cross-entropies, so neither trained arm’s cost has an interval and neither is claimed to differ from zero.

Two caveats bound this. The penalty worked asymmetrically: over the same 600 steps the *physics* gate became less selective in both arms, rising from 0.0110 to 0.0161 on constitution batches and 0.0113 to 0.0292 on no-harm ones, which is the term the penalty was pushing down. And the composition was measured on cross-entropy and accuracy only — no refusal comparison, no guard-rail suite on the dual stack — so “free” is established for what these two metrics see and not for behaviour.

What it says, and on what evidence. A frozen backbone will accept two latent channels at once, from partners of entirely different kinds, with no measurable interference and no joint training: the composition is a property of the interface, not something that has to be learned. That was measured on cross-entropy and on physics accuracy. Run on behaviour — the same guard-rail suite the single channels were held to, over the identical 100 cached items (Table 12) — it splits in two, and only one half of it holds.

Neither channel costs the other its payload. The values channel behaves in the dual stack almost exactly as it does alone. Red-team refusal is 0.710 against 0.720, red-team KL 0.1091 against 0.1047, and the five prompts that flip toward refusal with both channels open are a strict *subset* of the six that flip with the constitution channel alone — the two arms disagree on 1 of 100 red-team prompts — and adjudicated, each changes exactly one decision, the same prompt. Whatever the physics channel is doing, it is not displacing the constitution channel’s write, and it is not being displaced by it. That is the part of the composition claim that survives contact with behaviour, and it is the part the bit-identity properties above predict.

The composition is not inert on the backbone, which neither channel at its trained cap was. With both channels open MMLU divergence to base is 0.0723 with a p90 of 0.205 — sixteen times the 0.0045 the constitution channel alone produced, and the largest off-task perturbation anywhere in this paper. Accuracy, scored with the repaired parser, goes 75 to 77 (3:1, $p = 0.63$) against 1:1 for the constitution channel alone; the divergence is the finding, and the accuracy is not one — GSM8K and BoolQ stay where they were. So the claim that the two channels compose with *no measurable interference* is false as stated: it was true of cross-entropy and of physics accuracy, and it is not true of multiple-choice behaviour.

We are careful about which way to read this. The movement is one-directional and it is an *improvement*, so “interference” is the wrong word for what was measured — three items gained and one lost is not damage, and a reader who wanted the dual stack to be harmless gets that. What fails is the stronger and more interesting claim, that the composition is free: adding a second channel changed the backbone’s answers on a benchmark neither channel moved by itself.

Which channel, and the subject names are not the answer. The three gained items are `high_school_mathematics` and `college_physics`, which points at the physics channel — it is the one with a reason to engage a mathematics prompt. That inference does not survive Section 7’s 410-coordinate arm. *Two of these three items are gained there too*, by the constitution channel alone with no physics channel present at all; only `college_physics:94` is unique to the dual stack. So these MMLU items are fragile under any sufficiently strong write at this depth, the subjects they belong to are a property of which items are fragile rather than of which partner is attached, and a reading that assigned them to the physics channel would have been wrong.

What does separate the arms is the size of the divergence, and it orders by write strength rather than by channel count: MMLU KL to base is 0.0045 for the full-width constitution write at cap 0.2, 0.0244 for 410 coordinates at cap 0.522, and 0.0723 here, where two full-width writes at cap 0.2 land at two different depths. The dual stack is the largest perturbation of the three and the only one carrying two payloads, and we still cannot say from this run how much of its MMLU divergence is the physics channel’s own write: the gate the bench records is the *constitution* channel’s, so the physics gate’s activity on those prompts is not in the record. The arm that settles it runs the same suite with only the physics channel open (`results/dual/guardrail_phys.sh`), and it did: with only the physics channel open, MMLU KL to base is 0.0409 (p90 0.111), against 0.0045 for the constitution channel alone and 0.0723 for both, so most of the composition’s MMLU divergence is the physics channel’s own write, and the two together exceed their sum by 0.027. Its accuracy is unchanged (74 against 75, one item each way), GSM8K is item-identical at 83, BoolQ loses one item, and its red-team divergence is 0.0005: the physics channel has no behaviour of its own there.

What the values channel carries is the deflationary half of this paper, with one boundary now drawn more carefully than it was. A narrow write into probe-identified coordinates transmits a prompted teacher’s manner and not its judgment *at the perturbation budget the cap allowed it*; the width at which judgment did transmit is also the width that imported the teacher’s over-refusal; the coordinates were chosen on a causal claim that does not reproduce at either scale; and a partner that has memorised the document is interchangeable with — at nine dimensions, slightly worse than — one that has never read it, because the document enters through the teacher’s context rather than the partner’s weights. Three of those four are findings about this design. The first is, for now, a finding about how we configured it.

10 Discussion and Limitations

What has been shown. The coupling interface of Furui [5] accepts a partner that computes nothing and holds a document instead, and the resulting channel changes the backbone’s output in the direction of a teacher that has read the document. It does so while leaving arithmetic and multiple-choice reasoning within one or two items of the frozen backbone, and item-identical at 41 coordinates, with divergence two orders of magnitude larger on prompts the document speaks to than on arithmetic — $302\times$ at full width — and a gate whose own selectivity is highest where the write is narrowest, which is the behaviour the no-harm arm trains for and the property that makes such a channel auditable at all. Two channels of unrelated kinds compose on one backbone without joint training, neither costing the other its payload — though the composition is not inert

on the backbone, and Section 9 reports where it is not. And the probe that locates the write site reproduces at two scales, with a signal that occupies a roughly constant number of coordinates rather than a constant fraction.

What has not been shown, and what was withdrawn. The causal claim that motivated writing into value neurons does not reproduce: at 0.5B the damage is mostly magnitude, at 9B there is no damage to explain. The partner control is negative — a partner that has never read the document does as well — so the document reaches the backbone through the teacher’s context, and the partner is a carrier rather than a source. And judgment, adjudicated on substance rather than counted by keyword, arrives only at full width and only with enough prompts to see it: on 100 prompts no arm withholds more than the backbone at $p < 0.05$, one decision changes and it changes in every arm, and two arms give more on a prompt the backbone refused; on 400 the wide write withholds on 21 pairs against 2 ($p = 0.0001$), and what it withholds is, by two blind judges who agree on every pair, legitimate information fifteen times, dual-use material four times and harmful specifics twice. The judgment that arrives is a blunter refusal, and it is the document’s: a content-free injection of the same magnitude through the same gate changes 8 decisions one way and 4 the other ($p = 0.39$) at 5% of the divergence. The probe-best 410 at matched energy, given the same 400 prompts, withholds 16:5 ($p = 0.027$), thirteen of them legitimate information, and twice hands over harmful specifics the backbone had refused; the probe-worst set does the same, 15:2 ($p = 0.0023$), and releases the same precursor list, so that failure belongs to the 410-coordinate write at this energy and not to the probe’s coordinates. The narrower masks at parity, given those prompts, carry a weaker copy of the same withholding, 10:4 at 41 coordinates and 10:6 at 205, neither significant at $n = 400$, on largely the wide write’s prompts, and release nothing harmful. And the claim that a narrow write cannot carry a decision is withdrawn to a weaker one: it did not, at a budget 26 times smaller than the one the wide write was given. At matched energy the probe-best 410 carried most of the wide write’s fit and a weaker, non-significant copy of its keyword shift (1:4 away:toward, $p = 0.375$, on 100 items; Section 7); the 41- and 205-coordinate masks at matched energy carry half and 58% of its fit and, on 400 prompts, a withholding in the wide write’s direction that does not reach significance. So the honest summary of this paper’s central question — can a narrow, auditable write carry a disposition? — is that it carries a weaker copy of one, in the same direction and on largely the same prompts, at the MMLU cost the probe’s coordinates exact at every width (Section 8); whether that copy clears significance is a question of strength rather than of width, and at 41 coordinates the magnitude-matched control has decided that the cost is the coordinates and not the write: a random 41 written just as hard leaves every benchmark item-identical (Section 8). The 205 control is training as this is written.

The scaling conjecture, stated as a conjecture. If the value signal occupies a fixed number of coordinates as backbones widen (Table 3), and if a few hundred coordinates can carry a decision when given the budget to do so, then this kind of coupling gets *cheaper* in relative terms at frontier scale: the bridge’s parameters grow linearly in d_{model} , the mask costs nothing, and the fraction of the stream disturbed falls. Neither antecedent is established. The second is the subject of a running experiment; the first rests on two scales and a coarse grid. We also note a limit that the arithmetic makes visible: restoring energy parity with a full-width write by raising the cap scales as $\sqrt{d_{\text{model}}/n_{\text{mask}}}$, which at frontier width would demand per-coordinate writes far outside the residual stream’s own distribution. So parity with full width is the wrong asymptote to aim at. The quantity that governs whether this scales is the *minimum* energy that moves a decision, and nothing here has measured it — full width’s KL of 0.105 for a six-item refusal shift is very likely far above it.

Other limitations. Refusal was adjudicated on 400 held-out prompts by model judges against a written rubric, with a 40-pair independent re-judge per arm, on one seed, one machine and one backbone; the keyword count remains a proxy, and the two instruments largely disagree on which

prompts move (7 of the wide write’s 21 adjudicated withholdings are keyword flips). The teacher’s continuations at 9B were generated through a cached-prefill path whose identity check against an uncached decode passed 3 of 3 at 0.5B and **0 of 3** at 9B (the path is self-consistent, and the divergence is greedy ties under different prefill chunking, but every 9B number inherits the caveat). The constitution enters as a 2,848-word excerpt, not the whole document. All backbones are 4-bit quantized. And everything was run once, on one machine, at one seed.

11 Reproducibility

Everything is in <https://github.com/ryoji-info/PsiLM-2>, with the coupling interface and the physics-partner work in <https://github.com/ryoji-info/PsiLM>. The value-neuron probe, its magnitude controls, the constitution bridge at both scales, the width arms and the dual stack are driven by the shell chains under `results/`, each resumable and each writing the log the paper cites. Two verification scripts check the properties the dual stack’s attribution rests on: `psilm2/dual_self_test.py` (nine assertions, including three bit-identity properties and a gradient audit) and `psilm2/verify_qwen35.py` (the same three on the real 9B). Two more make claims checkable that were previously asserted from gitignored trajectory stores: `eval/vn_rms_full.py` writes the per-coordinate activation magnitudes behind every magnitude claim and every energy figure in Table 8, and `eval/const_paired_bootstrap.py` regenerates the paired intervals. `eval/const_refusal_mcnemar.py` regenerates the paired keyword tests, and `eval/const_refusal_adjudicated.py` the adjudicated ones from the per-arm labels under `results/constitution/adjudication_qwen35/`, each with the judge’s one-sentence reason and the re-judge’s label on the sampled pairs; the 400-prompt labels of the full-width, the two 410 and the two narrow parity arms are `const_qwen35_{all,vn10e,vn10ebot,vn1e,vn5e}_rt400.json` (four judges of 100 pairs each, a 40-pair re-judge), the content-free control’s are `const_qwen35_all_rt400_contentless.json`, and the two-judge categorisation of every changed pair is `results/constitution/withholding_categories*_rt400.json`, one directory up from the labels.

Two corrections from the development review, 2026-09-17. The letter parser had no bare-letter branch, and the 9B backbone answers 18 of 100 MMLU items as “B” or “(C)” then end-of-sequence; every one scored as wrong in every arm, so base MMLU read 64 where it is 75, and an arm that reformatted a few of those answers read as an accuracy gain. `eval/rescore_mmlu.py` regenerates every tracked summary with the repaired parser; every MMLU accuracy in this paper is the rescored one, and the KL columns, which are teacher-forced, never touched the parser. Separately, the teacher-forced KL path pools the forward readout over prompt *and* continuation where generation pools over the prompt alone; the discrepancy is common-mode across arms and small (about 2% of the KL mean on a small stack), and it is deliberately left in place until the queued arms finish so that every KL in the campaign shares one basis, after which it will be fixed and the shift quantified against the record. The runs in this paper — the constitution bridges at both scales, the Qwen3.5 physics bridge, the dual arms, and their evaluations and guard-rails — sum to about 177 GPU-hours on the one M2 by the measured rates in the run logs: the `psilm2` block of `results/compute_summary.json`, written by `eval/total_compute.py`, sums the `stage2c.*` and `stage2_qwen35` training logs (105 h), their 50-item evaluations (19 h) and the `const.*` and `dual_qwen35.*` guard-rails (52 h), excluding value-neuron state collection and probe training, data building, the partner’s fine-tune and the 0.5B ablations (the file’s unscoped totals also count the companion paper’s runs). The arms still queued at submission (the matched control at 205 coordinates and the 9B plain-partner twins) add about 30 more.

12 AI Generation Disclosure

The work reported here was carried out by Claude (Anthropic) — the Fable 5, Fable 5.1 and Opus 5 models, as recorded in the repository’s commit trailers — working on the author’s Apple M2 laptop under the author’s direction. Claude wrote the code, ran the experiments, chose most of the intermediate designs, and wrote this paper. The author set the research direction, approved or redirected each campaign, and decides what is published.

One part of the process is worth recording because it changed the content. Before publication the manuscript was audited by separate Opus 5 agents reading the repository against the text, which found sixteen claims that the logs did not support — two of them reporting bugs rather than wording — and the corrections are in the git history. The correction in Section 7 has a different origin: it came from the author asking whether a wider mask might work, which sent Claude back to the injection code to check what the cap actually bounded. Neither the original campaign nor its audit had questioned it.

Acknowledgments

The constitution text is Anthropic’s, released under CC0 [1]. The value-neuron probe follows Xu et al. [6]; the harmlessness prompts are `hh-rlhf` [3].

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